

GRADUATION PROJECT — DOCUMENTATION

Complaints Management System

Menofia Governorate

ITI — Open Source Application Development Track, Menofia Branch

2026

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1. Team Members

- Mahmoud Abdelhalim Ahmed Isa
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- Mahmoud Ismail Mohamed Elfiky
- Amir Ahmed Said Soliman

2. Project Idea & The Problem It Solves

The Problem

The Menofia Governorate (Egypt) handles citizen complaints through a **legacy Microsoft Access-based system**. This old system has serious limitations:

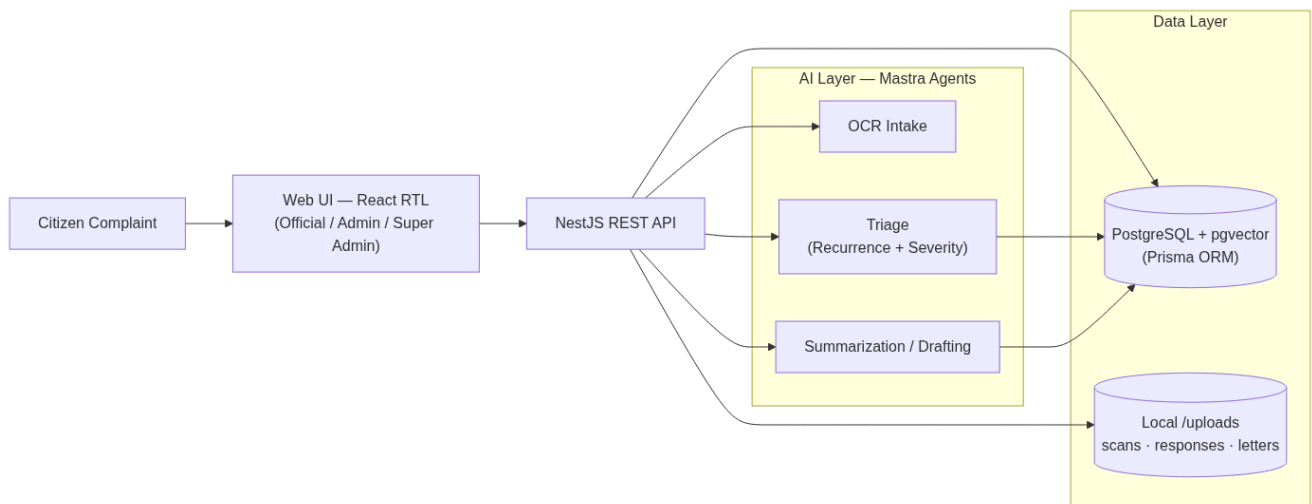
- Data entry is fully **manual** — every complaint must be typed by hand, including scanned forms.
- **No reporting** — achievement percentages, delay analysis, and periodic statistics are computed by hand or not at all.
- **No AI assistance** — nothing detects recurring complaints, scores severity, or summarizes records.
- Complaint handling depends on paper flows: scanned forms, authority responses, and notification documents are managed outside the system.

The Idea

Build a **full-stack web application** that replaces the Access system while preserving its proven data model and workflows — then goes beyond it with **AI-assisted intake, triage, and drafting**:

- Digital complaint lifecycle: intake → department assignment → examination tracking → archival.
- AI OCR intake: extract fields from **scanned and handwritten** complaint forms with per-field confidence scores, so officials only correct low-confidence fields.
- AI triage: detect **recurring complaints** (same citizen + category + location, or similar text) and score **severity** (Low/Medium/High) with manual override.
- Reporting: achievement % by department, delay reports, scheduled (daily/weekly) and on-demand reports, PDF/Excel export.
- Letter/memo generation with custom templates and letterhead settings.
- Facebook social media monitoring with a human-reviewed draft queue.
- Secure user management with a **three-tier role hierarchy** (Official / Admin / Super Admin), human-approved password reset requests, and in-app notifications.
- **Human-in-the-loop everywhere**: AI drafts and scores, and even password resets, are never final without a human's approval.

The Solution At a Glance



3. Target Users & Project Scope

Who Can Use This Project

User	Description	Permissions
Official	Government official who handles citizen complaints (default role)	Full complaint lifecycle (intake, assignment, examination tracking, archival), OCR intake, AI triage review with severity override, reports, letters/memos, social-draft review, notifications
Admin	Back-office administrator	Everything an Official can do, plus : manage Official user accounts (National IDs shown masked), approve password-reset requests for Officials, maintain reference data (departments, complaint types, reception methods), manage letter templates & letterhead settings, configure delay thresholds, manage monitored social sources
Super Admin	System-level administrator — seeded only, cannot be created by any user	Dedicated system administration area: manage all user accounts including Admin accounts , approve password-reset requests for Admin/Super Admin users, see full (unmasked) National IDs

Role rules enforced by the backend:

- Only Super Admin can assign the Admin role; **no one can create or assign a Super Admin account**.
- An Admin cannot delete or manage other Admin accounts.
- Password-reset authority escalates: Officials → Admin or Super Admin; Admin/Super Admin → Super Admin only.
- **Route protection uses exact role matching** — a route annotated Admin is *not* accessible to Super Admin unless explicitly listed. The Super Admin area is dedicated to system user administration; complaint operations and the Admin-only reference/settings screens are not exposed to it.

Project Scope

- Complaint lifecycle management (intake, assignment, examination status, archival)
- Citizen registry, complaint files, document storage
- Reporting (achievement %, delays, custom, scheduled + on-demand) with PDF/Excel export
- Letter & memo generation
- AI OCR intake (handwritten forms included)
- AI triage (recurrence + severity) with human override
- Delay tracking with admin-configurable thresholds
- AI summarization & drafting (human-approved drafts only)
- Facebook social media monitoring (official Graph API)
- Three-tier user management (Official / Admin / Super Admin) with National ID verification
- Human-approved password reset requests and in-app notifications

4. Technologies Used

Frontend

Technology	Purpose
React 19 + TypeScript + Vite	UI framework & build tool
Tailwind CSS + Radix UI	Styling & accessible components (RTL-aware)
React Router v7	Client-side routing + role guards
TanStack Query	Server state & caching
React Hook Form + Zod	Forms & validation
Recharts	Charts (achievement %, dashboards)
Axios	API communication

Backend

Technology	Purpose
NestJS 11 (TypeScript)	Modular backend framework
Prisma ORM 7 + PostgreSQL (+ pgvector)	Database & vector search
Passport + JWT + bcrypt	Authentication & password hashing
NestJS Schedule	Daily/weekly cron report jobs
ExcelJS	Excel (.xlsx) report export
Puppeteer / LibreOffice	PDF generation
Mammoth	DOCX template parsing/import

AI

Technology	Purpose
Mastra (agents, tools, workflows)	AI agent framework running inside the backend
ITI SBG gateway (Amazon Nova / Claude)	LLM access under the institute's data-handling agreement
Tesseract.js + Jimp	OCR engine + image preprocessing (Arabic & handwriting)
OpenAI text-embedding (via Mastra router)	Vector embeddings for complaint text similarity

Infrastructure / DevOps

Technology	Purpose
npm workspaces	Monorepo (apps/api, apps/web, packages/db, packages/types)
Docker (pgvector/postgres)	Local database
GitHub Actions (CI)	Continuous integration
Bruno	API integration tests (bruno:test / bruno:ci)

5. Features

Every feature is implemented end-to-end (backend API + frontend screen). Screenshots are stored in docs/screenshots/.

1. **Authentication & Three-Tier Role Access** — JWT login, role guards (Official / Admin / Super Admin), seeded admin + super admin accounts.

نظام إدارة الشكاوى

تسجيل الدخول للنظام

أدخل بياناتك للوصول إلى لوحة التحكم

البريد الإلكتروني أو اسم المستخدم

example@enterprise.com

كلمة المرور

تذكرني

تسجيل الدخول

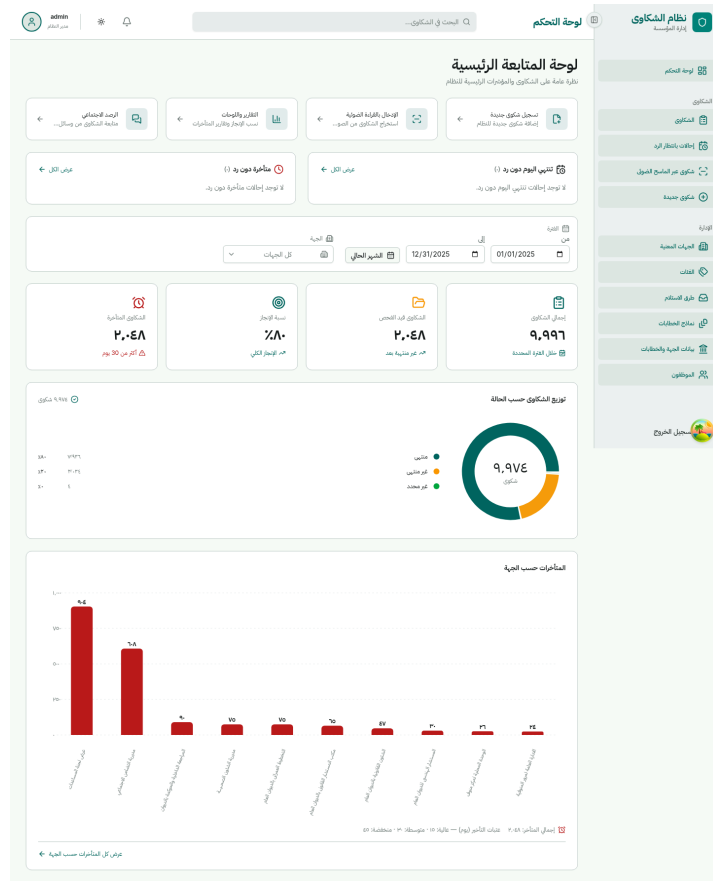
تواجه مشكلة في الدخول؟ [تواصل مع الدعم الفني](#)

الدعم الفني | شروط الخدمة | سياسة الخصوصية

نظام إدارة الشكاوى - جميع الحقوق محفوظة.

Screenshot: Login page

2. **Dashboard** — overall achievement %, open complaints, overdue counts, per-department breakdown.



Screenshot: Dashboard

3. Complaint Management — create, edit, list/search (partial name, ID+year), record navigation (First/Last/Next/Previous), department filter.

إدارة الشكاوى						
متابعة ومعالجة جميع الشكاوى الواردة						
رقم الشكاوى	الموضوع	المواطن	القسم	الأولوية / الحالة	تاريخ الإنشاء	الإجراءات
#8775-2026	شراء هاتف محمول معيب	أحمد محمد علي	مديرية التموين والتجارة الداخلية	متوسطة	٢٠٢٦/٨/١٢	...
#8774-2026	عيب في شاشة هاتف محمول	أحمد محمد علي	الوحدة المحلية	متوسطة	٢٠٢٦/٨/١٢	...
#8773-2026	In tempore ut dolor	محمد عبد الرزاق احمد مصطفى الجمل	مديرية البيئة	متوسطة	٢٠٢٦/٨/١٠	...
#8771-2026	قمت بشراء هاتف محمول من فرع مكتبة الـ...	أحمد محمد علي	-	متوسطة	٢٠٢٦/٨/٢	...
#7921-2026	طلب صيانة خط مياه في قرية كفر الدوار	أحمد محمد علي	-	متوسطة	٢٠٢٦/٧/٢٧	...
#7920-2026	طلب صيانة خط مياه في قرية كفر الدوار	أحمد محمد علي	-	متوسطة	٢٠٢٦/٧/٢٧	...
#7919-2026	طلب صيانة خط مياه في قرية كفر الدوار	أحمد محمد علي	-	متوسطة	٢٠٢٦/٧/٢٧	...
#1619-2026	صرف مساعدة مالية نظرا لمرض ابني وعلى...	اميرة عادل ابراهيم المليحي	عرض لجنة المساعدات	متوسطة	٢٠٢٦/٧/٢٥	...
#2958-2026	طلب الحصول على مساعدة مالية لمواجبه...	ليلى سعيد عشري الجزائر	عرض لجنة المساعدات	متوسطة	٢٠٢٦/٧/٢٥	...
#2679-2026	طلب مساعدة حيث ان ابنته تعاني من اتيب...	السيد عبدالعزى محمد دويدار	عرض لجنة المساعدات	متوسطة	٢٠٢٦/٧/٢٥	...

Screenshot: Complaints list

Screenshot: New complaint form

4. Complaint Detail & Lifecycle — examination status, severity badge, recurrence links, files, AI summary.

Screenshot: Complaint detail

5. Archiving, Reassignment, Responses & Urgency — full examination workflow.

admin | مدير النظام

البحث في الشكاوى...

نظام الشكاوى
إدارة المؤسسة

لوحة التحكم

الشكاوى

الشكاوى

إحالات بانتظار الرد

شكاوى عبر الماسح الضوئي

شكاوى جديدة

الإدارة

الجهات المعنية

الفتات

طرق الاستلام

نماذج الخطابات

بيانات الجبة والخطابات

الموظفون

سجل الخروج

أرشفة الشكوى

#7920-2026 — طلب صيانة خط مياه في قرية كفر الدوار

رقم الحفظ *

رقم حفظ الأرشيف

موقع الحفظ

تاريخ الأرشيف

موقع الحفظ (اختياري)

mm/dd/yyyy

أرشفة

إلغاء

Screenshot: Archive/Reassign/Response

6. Citizen Registry — National ID lookup, name search, village/location hierarchy.

admin | مدير النظام

البحث في الشكاوى...

نظام الشكاوى
إدارة المؤسسة

شكاوى جديدة

لوحة التحكم

الشكاوى

الشكاوى

إحالات بانتظار الرد

شكاوى عبر الماسح الضوئي

شكاوى جديدة

الإدارة

الجهات المعنية

الفتات

طرق الاستلام

نماذج الخطابات

بيانات الجبة والخطابات

الموظفون

سجل الخروج

تقديم شكوى جديدة

يرجى تعبئة التفاصيل أدناه لمساعدتنا في معالجة شكاواك بفعالية.

1 بيانات المواطن

2 بيانات الشكاوى

3 المرفقات

4 المراجعة

معلومات المواطن

الرقم القومي

12345678901231

رقم الجوال

رقم الجوال (اختياري)

الاسم الكامل *

الاسم الكامل للمواطن

العنوان

العنوان (اختياري)

القرية

المركز

اختر المركز أولاً

اختر المركز

شكاوى المواطن السابقة

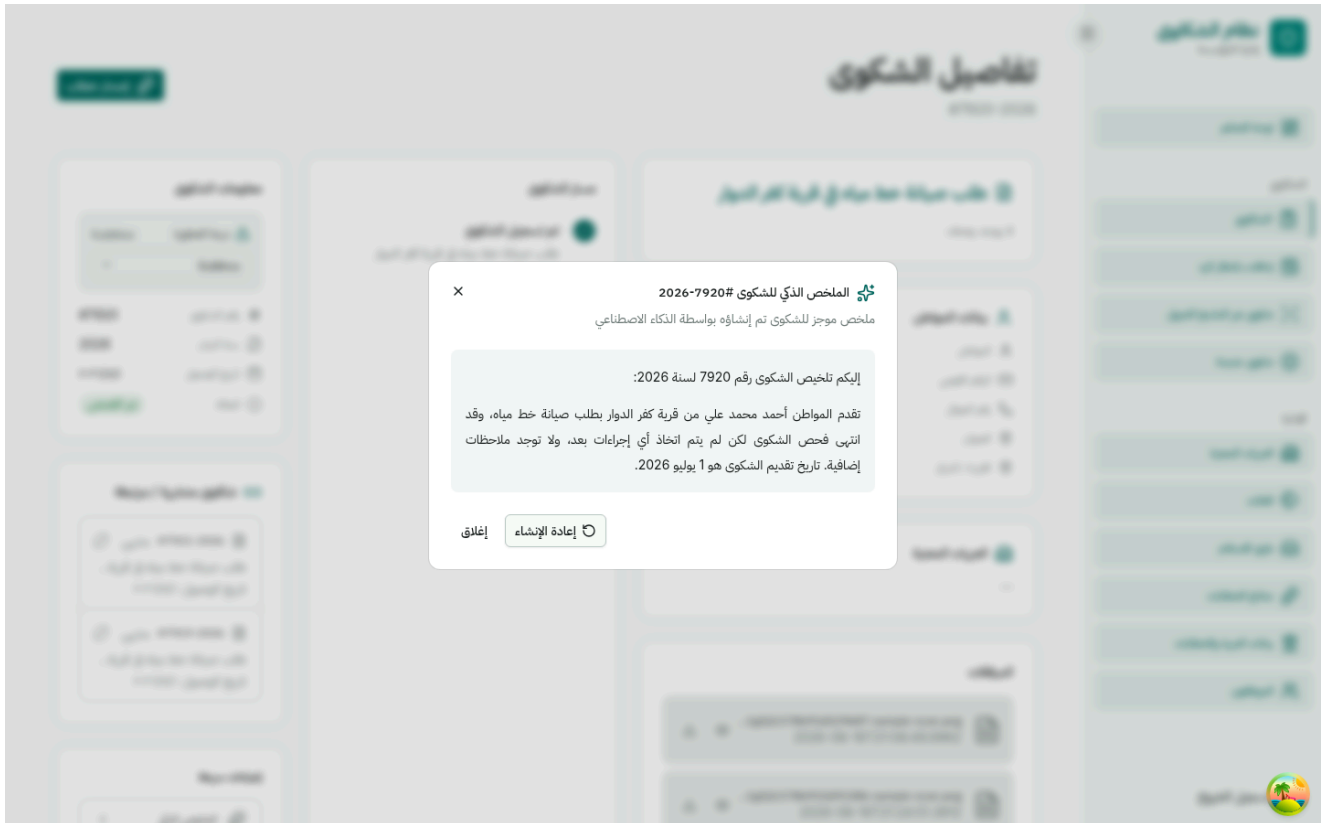
اعرض الشكاوى المسجلة مسبقاً لنفس الرقم القومي أو بنفس الاسم لتجنب تكرار التسجيل.

التالي

Screenshot: Citizen lookup

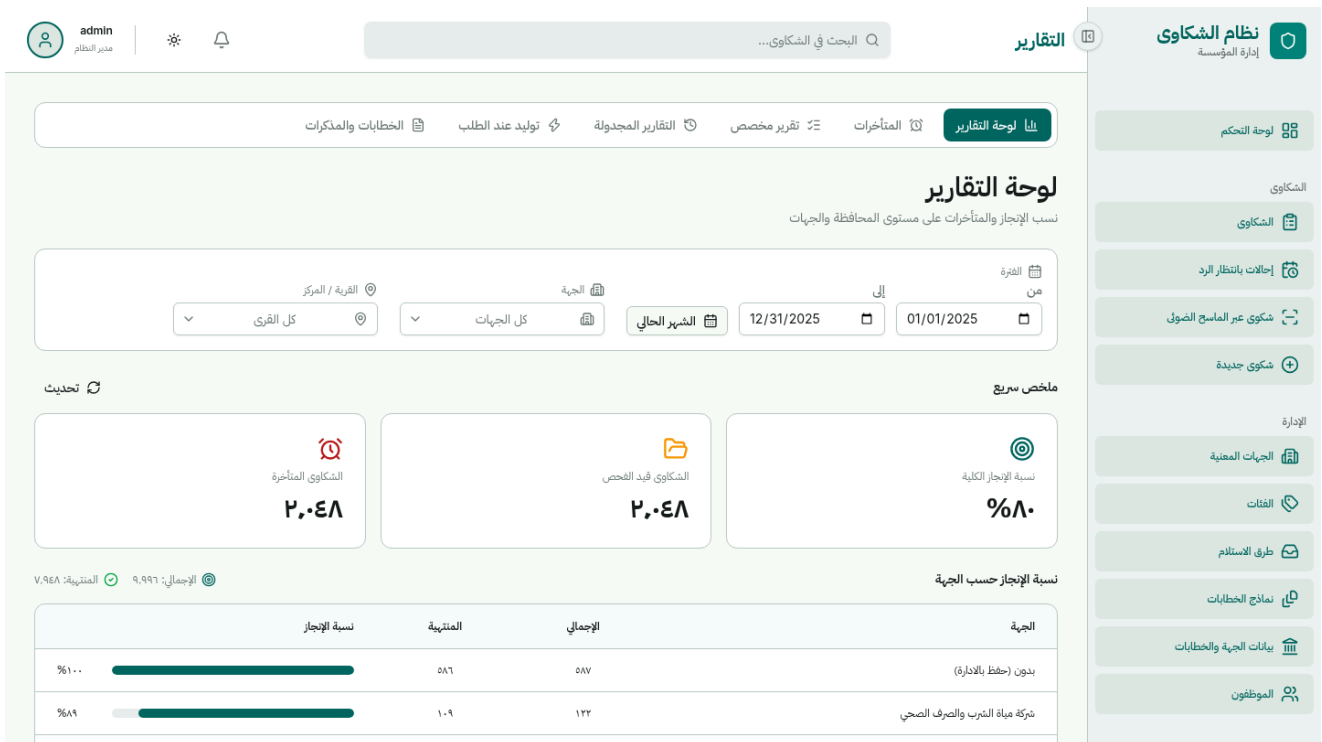
- File Attachments** — scanned forms, authority responses, notification documents (local /uploads volume).
- OCR Intake** — upload scanned/handwritten form → auto-filled fields with confidence scores → low-confidence fields highlighted for review.

9. **AI Triage** — recurrence detection with linked complaint history + severity scoring (Low/Medium/High) with official override.
10. **AI Summarization & Drafting** — complaint summaries, drafted reports and memos — all reviewable before finalizing.

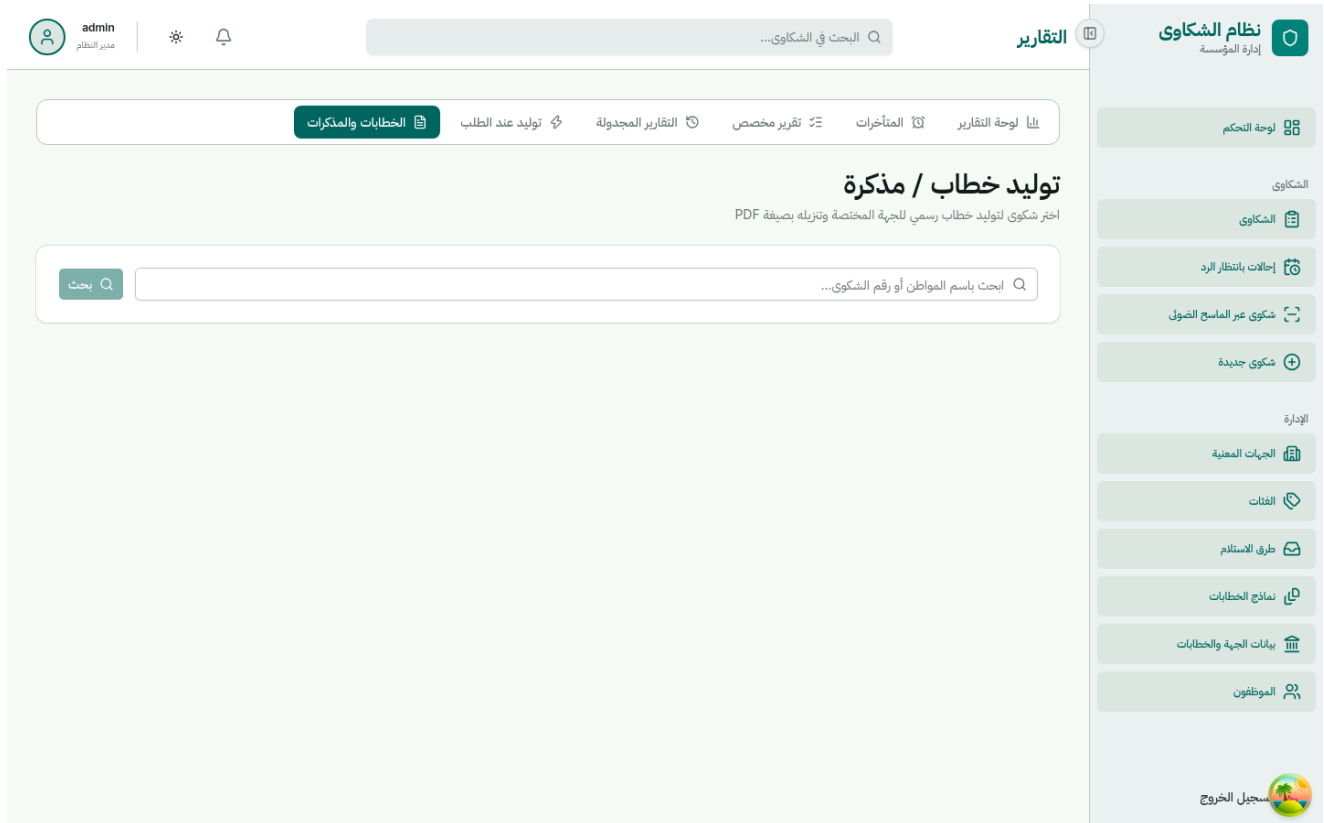


Screenshot: AI summary on complaint detail

11. **Reporting** — achievement % by department, delay reports, custom filtered reports, scheduled (daily/weekly) + on-demand, PDF/Excel export.

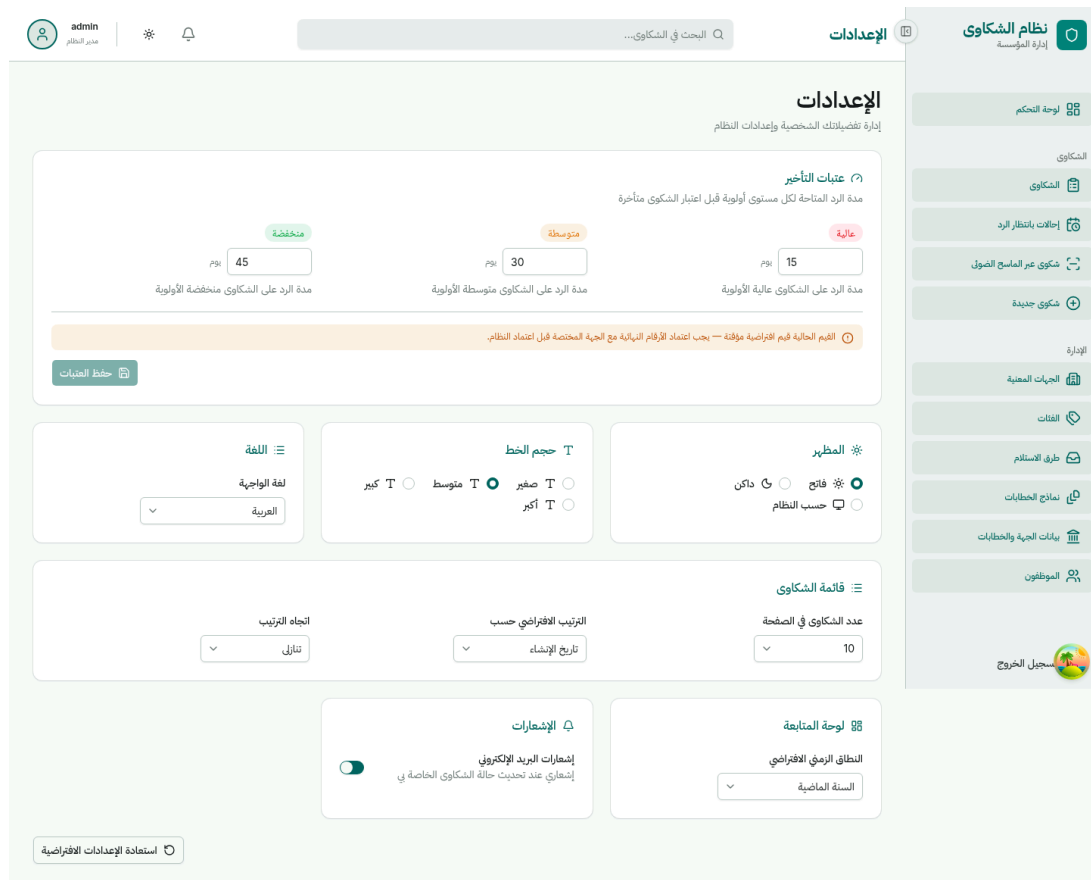


Screenshot: Reports dashboard



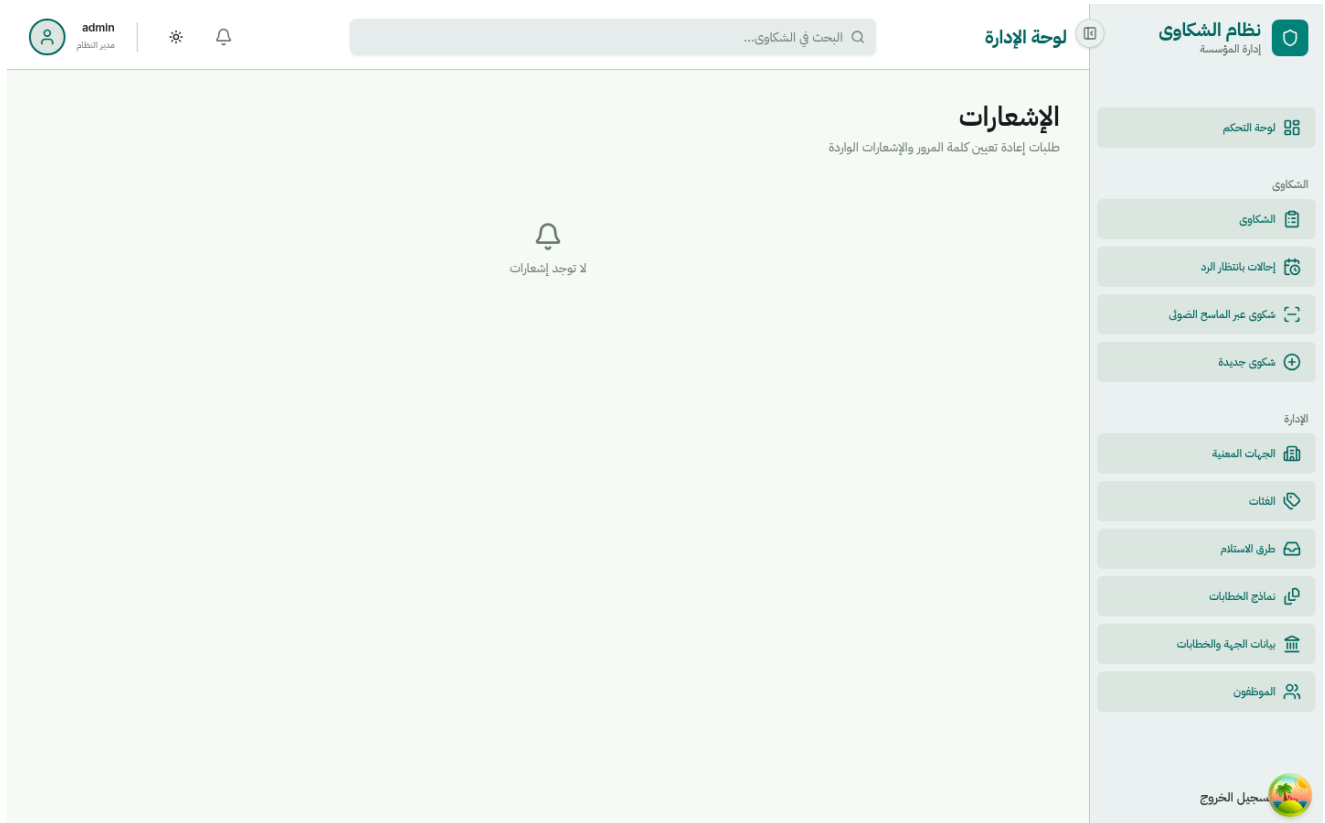
Screenshot: Memo generation

13. Delay Tracking — admin-configurable thresholds per severity (High/Medium/Low) and overdue visual flags.



Screenshot: Delay thresholds settings

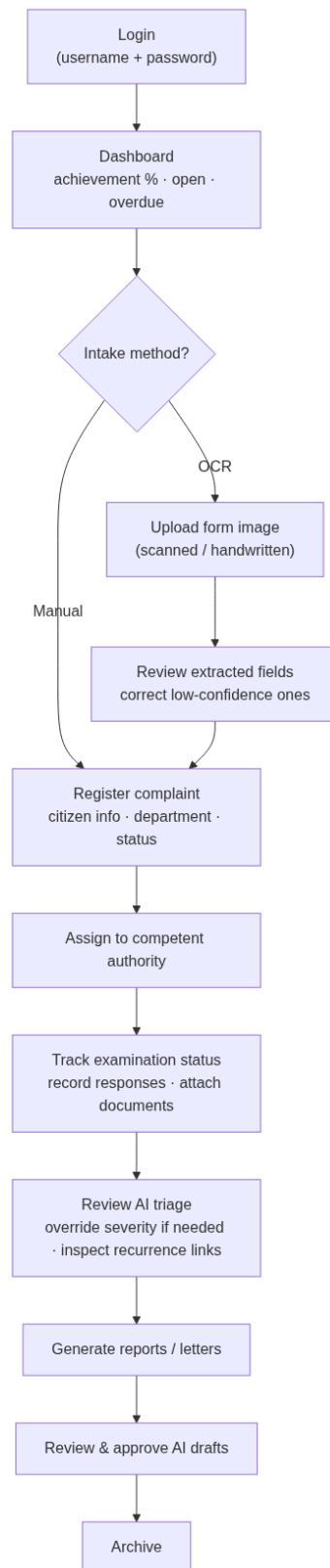
14. Social Media Monitoring — Facebook governorate page + community groups via official Graph API; flagged posts become draft complaints for human review.



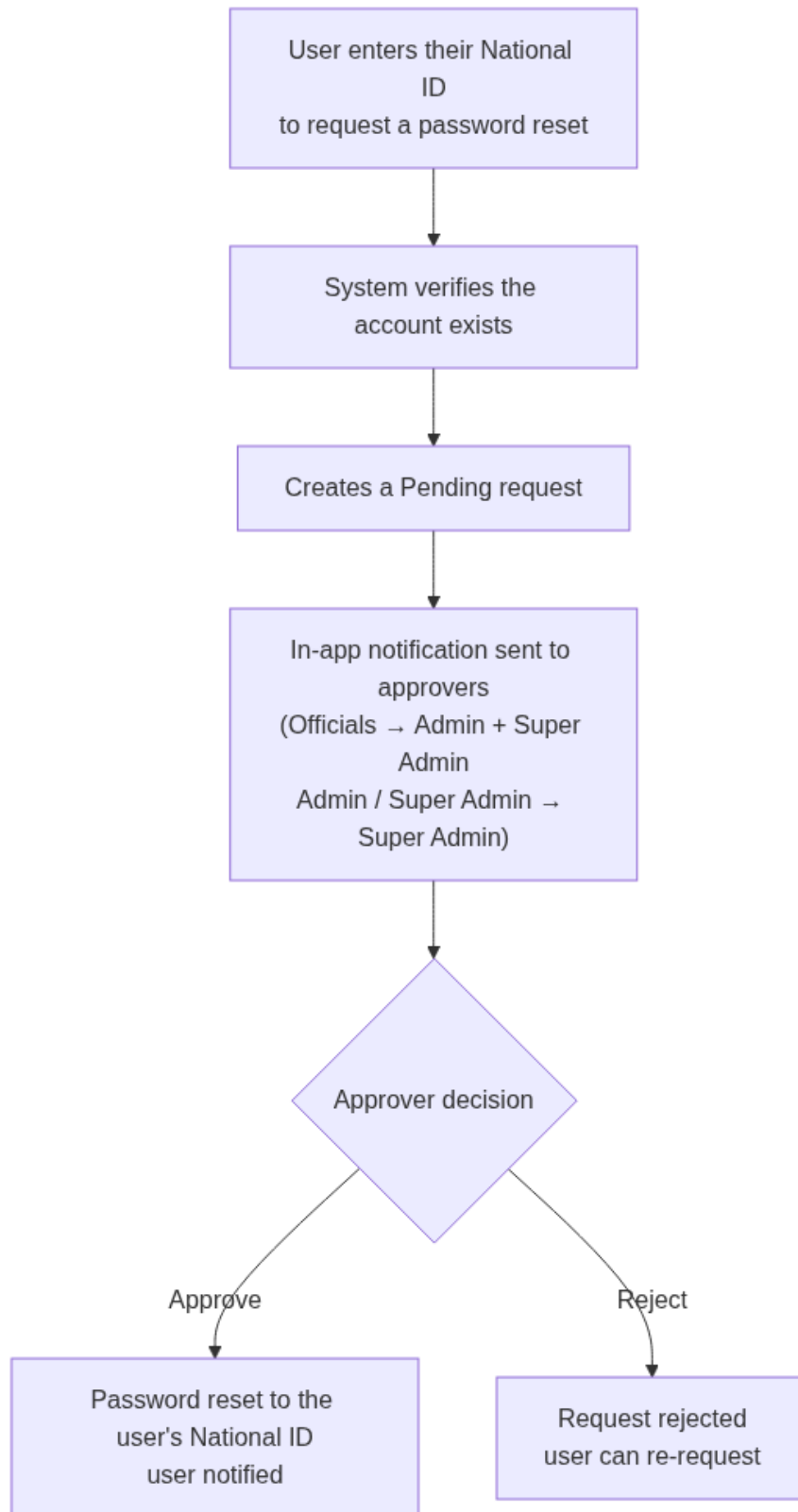
Screenshot: Notifications

6. Users & Usage Flow

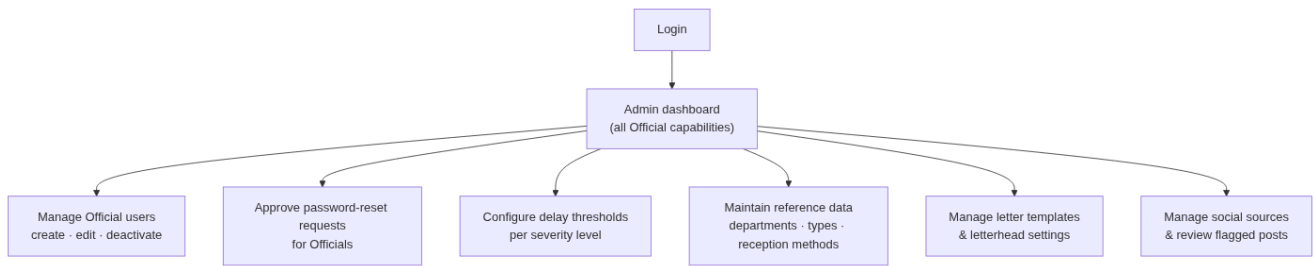
Official Flow



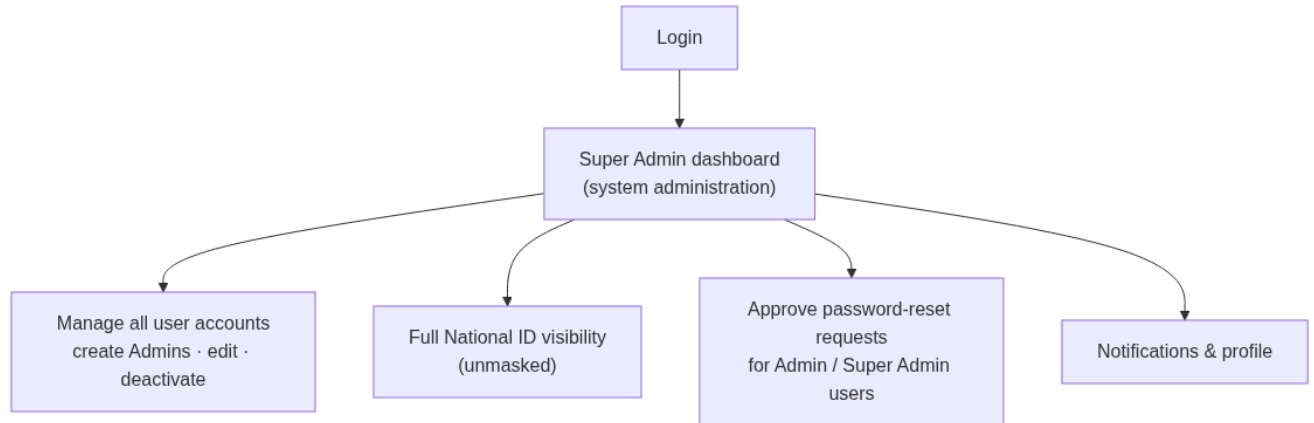
Password Reset Flow (any user)



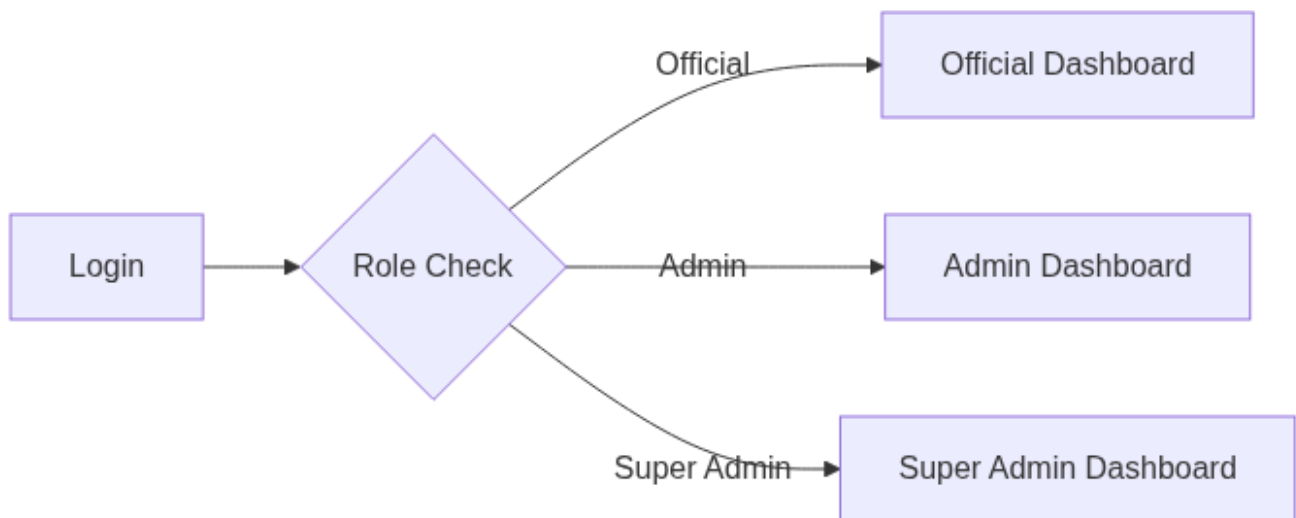
Admin Flow



Super Admin Flow



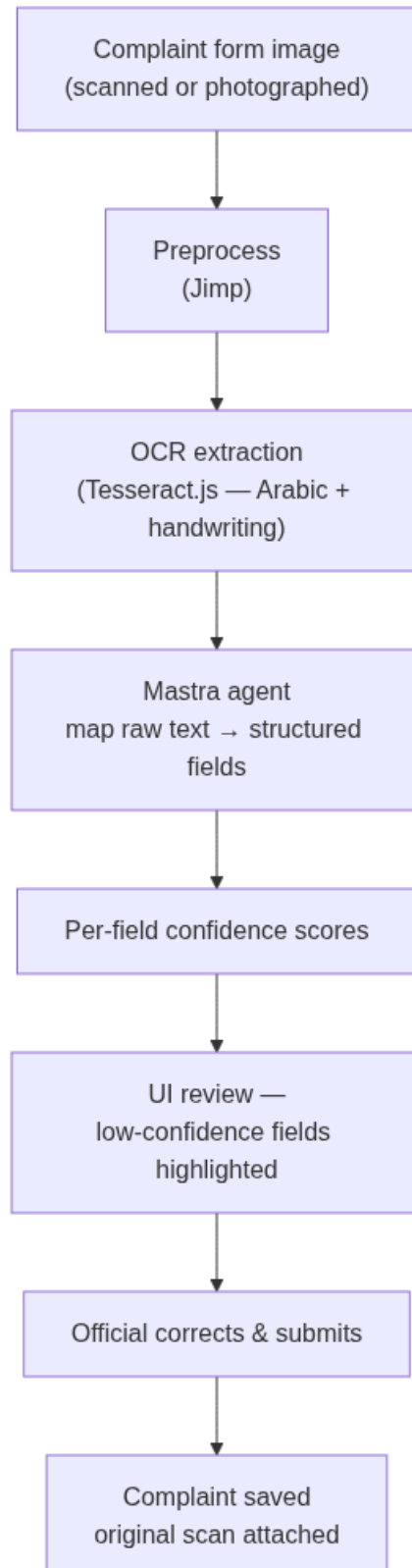
Flow Summary



7. AI Integration

All AI features are implemented as **Mastra agents/tools running inside the NestJS backend**, over an LLM accessed through the ITI SBG gateway (with the institute's data-handling agreement). Everything AI produces is a **draft or suggestion** — never final without a human.

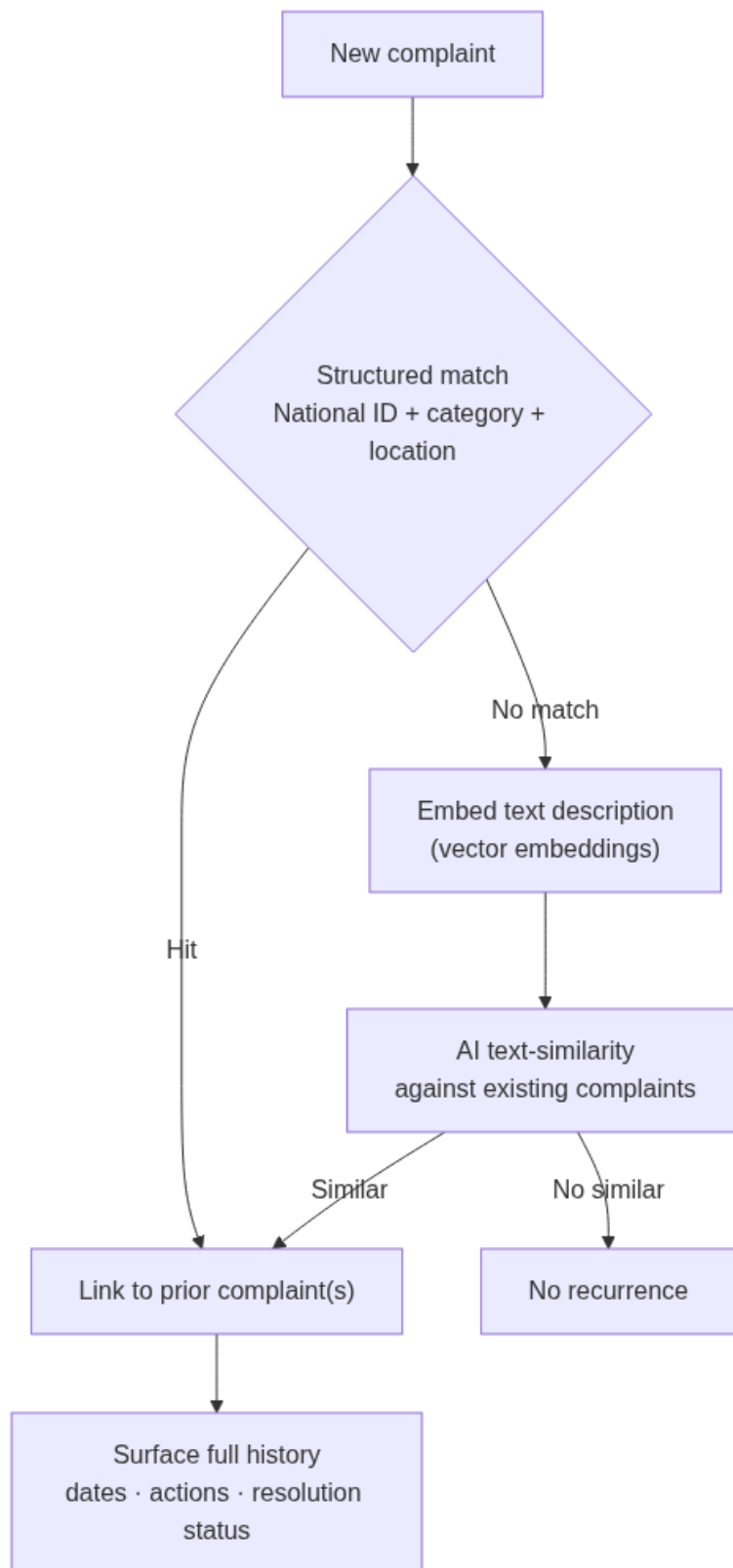
7.1 OCR Intake (Phase 2)



- The UI visually highlights low-confidence fields so the official knows what to check.
- The original scan is stored and attached to the complaint.

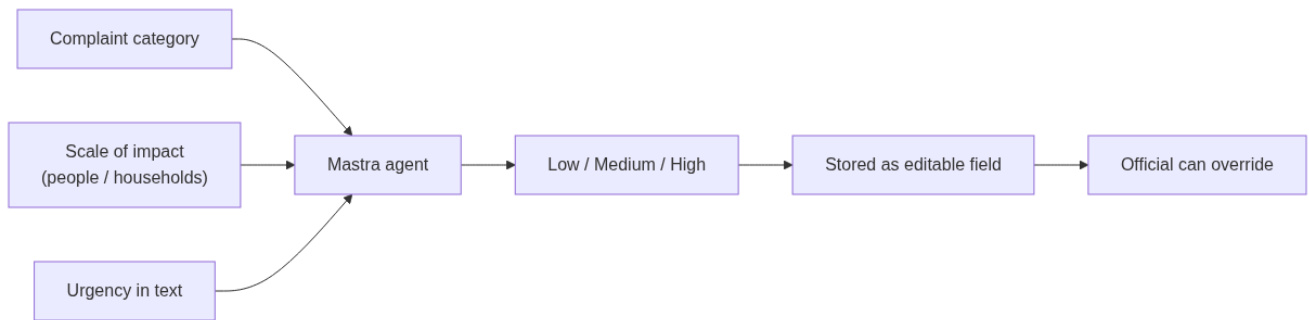
7.2 Recurrence Detection (Phase 3)

Layered matching — structured first, AI as fallback:



- Catches cases where a *different* citizen reports the *same* underlying issue.

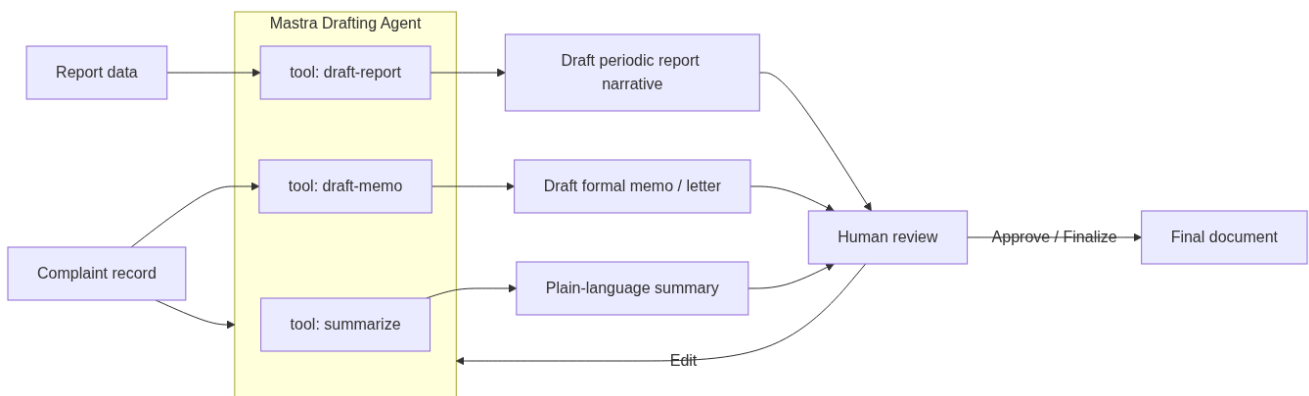
7.3 Severity Scoring (Phase 3)



Severity is **visible and overridable** by officials — never locked as AI-only.

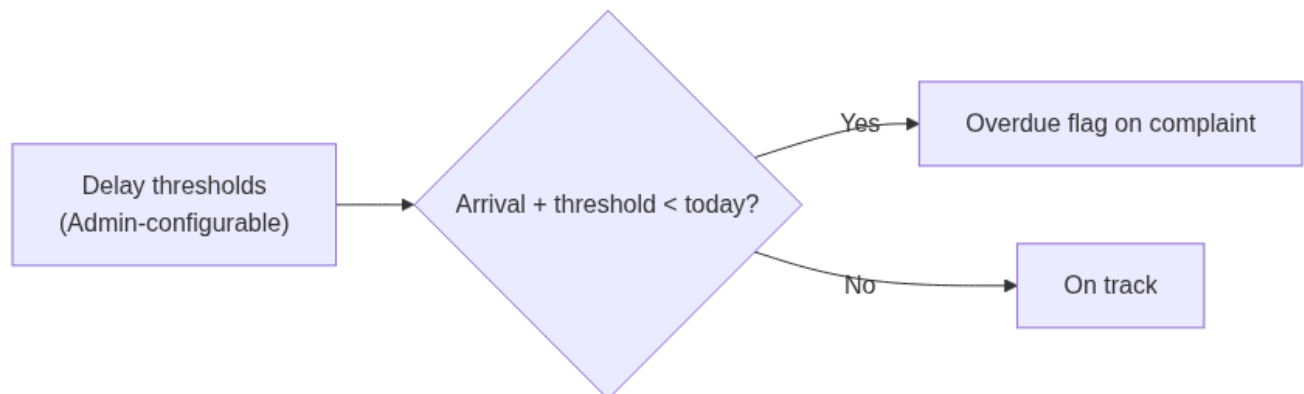
7.4 AI Summarization & Drafting (Phase 3)

One Mastra agent with three tools:



All outputs are **drafts only** — an “Approve/Finalize” action must be triggered by a human.

7.5 Delay Tracking (Phase 3, AI-adjacent)



Admin-configurable day thresholds per severity level (defaults: Low 45 / Medium 30 / High 15 days). Overdue complaints get a **visual flag only** — no auto-escalation.

8. Future Enhancements

Planned for the next versions:

- **Audit trail** — field-level history of every change
- **Automatic escalation & proactive notifications** — for high-severity or overdue complaints (currently: visual flags only, plus in-app notifications for the password-reset workflow)
- **More social platforms** — beyond Facebook
- **Mobile application** — for field officials
- **Historical data migration tooling** — import legacy Access records

9. Conclusion

The Complaints Management System replaces a dated Microsoft Access-based application with a modern web platform that preserves the governorate's proven workflows while adding AI-assisted intake, triage, and drafting. Built on a modular NestJS backend, a React frontend with an RTL Arabic interface, and a Prisma-backed PostgreSQL database, the system delivers the full complaint lifecycle, reporting, OCR-based intake, AI triage, human-approved AI drafting, a three-tier role hierarchy, and a secure password-reset workflow — a practical, production-oriented replacement for daily complaint operations.

10. References & Links

Item	Link
GitHub Repository	https://github.com/OS46-GP/complaints-system
Task Tracker / Jira	https://mahmoudhalim.atlassian.net/jira/software/projects/CS/boards/6